
Marine Coders Publication 1



Marine Innovation Unit

DISCLAIMER

This is not an official USMC publication. This is meant for internal use within the Marine Coders team and throughout the broader Marine Innovation Unit as necessary.

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Forward

This publication is designed to guide current and future Marine Coders in understanding their role, responsibilities, and impact. This document will be updated as situations change.

The Necessity of Software in the Marine Corps

The modern battlefield is digital, and nearly every aspect of modern combat operations relies on software in some capacity. From command and control (C2) systems to logistics management, software underpins the ability to fight and win. Consider the following examples:

- Precision guided munitions like JDAMs and GMLRS rely on software for targeting and guidance.
- Tactical data networks like Link-16 and Blue Force Tracker leverage software for real-time battlefield awareness and coordination.
- GCSS-MC (Global Combat Support System-Marine Corps) manages supply chains, vehicle readiness, and sustainment planning—all software-driven.
- The use of unmanned aerial vehicles (UAVs), robotic ground systems, and AI-powered decision support tools is increasing.

Who are The Marine Coders

Marine Coders are the digital warfighters of the Marine Corps, bringing world-class software development skills to enhance operational effectiveness and modern warfighting capabilities.

As a part of the Marine Innovation Unit (MIU), Marine Coders leverage the vast expertise of Marine Reservists who have excelled in the technology sector, bringing cutting-edge innovation, industry best practices, and real-world experience to Marine Corps modernization efforts.

Relevance to Force Design

The Marine Corps' Force Design initiative is a transformative effort to reshape the force to meet the demands of the future fight. As the modern battlefield becomes increasingly digitized, the ability to develop software at the tactical edge is no longer a luxury—it is a necessity. Marine Coders directly support Force Design by enabling rapid, adaptive, and secure software solutions that enhance warfighting capabilities across all domains.

Force Design envisions a Marine Corps capable of operating within contested maritime spaces, leveraging distributed forces, and integrating multi-domain reconnaissance and counter-reconnaissance to gain a decisive advantage. To achieve this, the force must be equipped with software-driven capabilities that provide:

- Real-time data processing and decision support to enhance operational command and control.
- Tactical AI/ML applications to improve battlefield awareness, automate threat detection, and optimize logistics.
- Cyber resilience and zero-trust security architectures to safeguard digital infrastructure in contested environments.

- Interoperability with joint and allied forces through seamless software integration and common data environments.
- Edge computing and cloud-based solutions to ensure connectivity and operational effectiveness even in degraded conditions.

The war in Ukraine and other recent conflicts have highlighted the importance of software-driven lethality and resilience. Digital applications have been used to coordinate fires, enhance targeting precision, and facilitate agile logistics—capabilities that Marine Coders will bring to the fight.

Through their work with Marine Corps Software Factory (MCSWF) and other Marine Corps stakeholders, Marine Coders ensure that the Marine Corps remains at the forefront of digital warfighting. The ability to develop, deploy, and sustain software solutions in a rapidly evolving battlespace will be a defining feature of 21st-century warfare, and Marine Coders are leading this charge.

Chapter 1

The History of the Marine Coders

“War is both timeless and ever changing. While the basic nature of war is constant, the means and methods we use evolve continuously.”

- MCDP 1, Warfighting

“The Marine Corps must become a more agile, resilient, and technologically advanced force. We must modernize—not for the sake of change, but to meet the demands of the future operating environment.”

- Force Design 2030

The story of the Marine Coders is a story of initiative, grassroots-energy, and impact.

The team was created in 2020 by two Marines—Captain Collin Chew and Major Drew Hutcheon—who saw something the Marine Corps lacked: organic software capability. While the force prided itself on being agile, expeditionary, and lethal, its approach to software lagged behind, often constrained by bureaucracy, compliance-first thinking, and reliance on external contractors. The Marines were becoming increasingly dependent on digital systems, yet the Corps had few warfighters who could actually build or fix the software they depended on.

So, rather than wait for a formal directive or funding line, these two officers started a community.

They called it Marine Coders—a grassroots, bottoms-up initiative built not from a staffing document, but from a shared passion. It began informally, as a community of interest, with Marines from across the Total Force—active duty, reservists, enlisted and officers—connecting through chat servers, Git repositories, and weekend code sprints. Many had backgrounds in computer science, engineering, or data, but no formal place to apply those skills within the Marine Corps. Marine Coders gave them that place.

What made Marine Coders unique wasn't just the technology it produced—it was the way it was built. It wasn't born out of a tasker or a Program Objective Memorandum (POM) cycle. It emerged organically, driven by Marines who saw that coding could be a form of warfighting. And through that lens, they began delivering tools that mattered—apps used in real operations, software that replaced spreadsheets and PowerPoint, and tools tailored by Marines, for Marines.

Key Milestones and Dates

September 2020: Marine Coders begins after Major Hutcheon and Captain Chew complete a tour to the USAF Platform One. They create the Marine Coders as an activity duty community of practice to align active duty talent across the enterprise and share best practices on developing software. Initially, it was modeled on the (now defunct) “Airman Coders” and “Supra Coders.”

October 2020: Though not “officially” sanctioned as a community of practice, the initial concept is presented to Lieutenant General Glavy (then Major General), who later becomes a champion of the Marine Corps Software Factory while serving as Deputy Commandant for Information.

September 2021: Major Hutcheon and Captain Chew meet Brigadier General Mahlock. She provides them executive sponsorship to build a “microservices challenge.” This is the first executive sponsorship to a Marine Corps wide hackathon.

October 2021: The II MIG Marine Coders 'Detachment' is formally chartered, marking an early effort to explore organic software development within the Marine Corps. However, the initiative sees limited operational engagement and is eventually sunset without sustained activity.

November 2021: Lieutenant Colonel Carlson, one of the founders of the MIU, connects with Major Hutcheon and Captain Chew as they are resigning from active duty. The three Marines immediately saw the alignment between the newly formed MIU and the Marine Coders team. Soon thereafter a formal announcement is released establishing the partnership.

XYZ 2021: Lieutenant Colonel Charlie Bahk is inspired to create the Marine Corps Software Factory (MCSWF) after a visit to the Army Software Factory.

August 2021: MIU's first annual training.

XYZ 2022: MIU's second annual training.

XYZ 2023: MIU officially stands up as a unit. There are ~15 Marine Coders at this time.

June 2024: The Marine Coders attend their first annual training in support of the MCSWF. There are ~20 Marine Coders in attendance.

March 2025: The Marine Coders attend their second annual training in support of the MCSWF. There are ~30 Marine Coders in attendance.

THIS SECTION NEEDS TO BE UPDATED WITH MCSWF AND MIU MILESTONES AND DATES

Chapter 2

Mission, Vision, and Core Values

“It must be considered that there is nothing more difficult to carry out, nor more doubtful of success, nor more dangerous to handle, than to initiate a new order of things.”

- Niccolo Machiavelli, The Prince

“There are two thousand years of experience to tell us that the only thing harder than getting a new idea into the military mind is to get an old idea out.”

- Basil H. Liddell Hart, Thoughts on War

Mission

To design, develop, and deploy software solutions that enhance the lethality, agility, and effectiveness of the Marine Corps

Vision

To transform the Marine Corps' software philosophy to a warfighting-first mindset that prioritizes lethality, agility, and operational effectiveness over acquisition cycles, centralization, and bureaucracy. Our vision is a Marine Corps where software is not just a support function but a combat multiplier—where Marines at the edge have the capability to build, modify, and deploy mission-critical software solutions in real-time.

Core Values

We are Marines first—regardless of our technical specialty or billet. Our work is grounded in the Marine Corps' core values of honor, courage, and commitment. As we lead in digital modernization efforts, we do so with a warfighting mindset, understanding that our software solutions must be as disciplined, reliable, and resilient as the Marines who depend on them. We view technology as a force multiplier, but never as a substitute for the ethos that defines us.

We are mission focused - Our development efforts are driven by operational relevance and tactical necessity. We write code with a purpose: to solve real problems for real Marines. Whether it's a planning tool for a MEU, a data dashboard for logistics, or a TAK plugin for a unit in the field, every line of code we write is intended to enhance the Marine Corps' ability to shoot, move, and communicate.

We measure success not in features delivered, but in capability gained by the warfighter.

We are agile and adaptive - We embrace the uncertainty of the modern battlefield by building software and teams that are agile, responsive, and decentralized. We deliver in small, rapid increments—listening to user feedback, iterating quickly, and deploying improvements with minimal friction. Our operational environment demands speed, flexibility, and innovation under pressure, and we apply those same principles to the way we build and ship software.

We are continuous learners - Technology evolves rapidly, and so must we. We are committed to lifelong learning—staying current with modern frameworks, tools, and practices so we can deliver high-quality software in a dynamic threat environment. Through asynchronous training, peer collaboration, and real-world problem solving, we invest in our individual growth and the collective capability of the team. We view every challenge as an opportunity to learn and improve.

Chapter 3

Marine Coders Organization and Structure

“Innovation distinguishes between a leader and a follower.”

- Steve Jobs

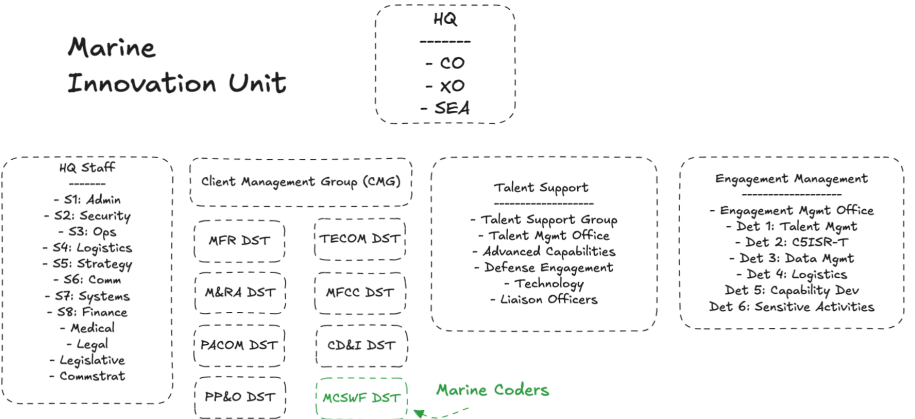
“Software is eating the world.”

- Marc Andreessen

Marine Coders operate as part of the Marine Innovation Unit (MIU), wherein they serve as a Direct Support Team (DST) to the Marine Corps Software Factory (MCSWF).

The MIU is a recently formed unit and, as such, their organizational structure is still evolving. In general, the staffing model of the MIU depends on the DSTs who reside in the Client Management Group (CMG). These DSTs maintain standing relationships with select organizations within the Marine Corps (e.g. PACOM, TECOM, MCSWF). The DSTs identify needs for their respective organizations and scope those needs into engagements which can be staffed and supported by members of the MIU. Once the engagements are approved by MIU leadership, they are staffed by the talent support division and executed by the engagement management division.

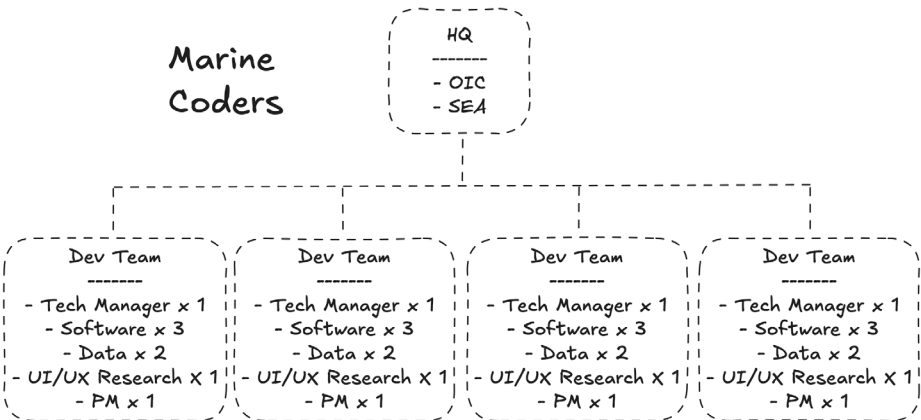
The Marine Coders are unique within this MIU operating model in that they identify MCSWF needs and staff the engagements to satisfy those needs themselves.



Marine Coders Composition

The Marine Coders development teams are designed for flexibility, incorporating a diversity of rank, skill sets, and technical expertise to ensure mission-critical software solutions can be developed efficiently. Unlike traditional Marine Corps units, where rank often dictates leadership roles, Marine Coders teams operate with a meritocratic, skill-based leadership approach, ensuring that technical expertise, product vision, and operational need drive decision-making.

Administratively, the Marine Coders team is comprised of four cross functional development teams. These teams are foundational to how Marine Coders recruits, screens, trains, and performs administrative and logistical functions. If possible, the integrity of these teams may be preserved for development efforts in support of customers. It is worth noting, however, that this team structure is not forced on problems. The Marine Coders are flexible in staffing individual teams to meet unique challenges for customer needs. In practicality, this means that development efforts can be staffed by select persons within administrative teams, or across administrative teams, depending on the specific skillsets needed.



Marine Coders Capabilities

The Marine Coders team maintains five core capabilities, listed below (note: order is not meant to indicate priority).

Tactical Application Development (TAK-Specific)

- TAK Plugin Development – Building and extending ATAK and WinTAK plugins
- Geospatial & Mapping (GIS) – Working with geospatial data for geospatial overlays
- UI Design – Understanding military UI/UX constraints and user workflows in the field

Web Development

- Building modern, scalable, and interactive front-end applications
- State Management - Handling complex application states
- Leveraging component library & UI frameworks for consistent UI/UX
- Integrating live data feeds

Backend Development

- Building modular, scalable backend applications
- Designing efficient API structures for TAK and web apps
- Secure access control for military applications
- Database Management

DevSecOps & Deployment

- Ensuring apps are deployable across environments
- Automating builds, testing, and deployment
- Managing deployment configurations efficiently
- Tracking performance and system health

Agile & Cross-Functional Collaboration

- Working in scrum or Kanban environments to deliver iteratively
- Engaging with customers to ensure applications meet real-world mission requirements
- Maintaining clear, well-documented code and processes for maintainability
- Ensuring compatibility with other DoD systems

Roles and Responsibilities

There are five primary roles on Marine Coder's development teams. A high-level overview of the roles and their responsibilities follows:

Technical Managers – Lead the design, development, and delivery of software solutions that support mission-critical needs. They guide architecture decisions, mentor junior developers, and ensure code quality and security across the development lifecycle. Technical Managers translate operational requirements into scalable systems and oversee implementation of modern development practices. Their leadership enables reliable delivery in support of mission objectives.

Software Engineers – Develop mission-critical applications using modern frameworks and DevSecOps principles. They write and maintain secure, scalable, and high-performance code to support warfighting operations, logistics, and intelligence functions. Their expertise includes cloud-native development, containerization, and full-stack application development.

Data Analysts & AI/ML Specialists – Process and analyze data to enhance battlefield awareness and decision-making. They build data pipelines, apply predictive analytics, and develop models for use in

applications and/or operational planning. Their work supports real-time data fusion for tactical and strategic decision-making.

UI/UX Researchers – Conduct user-centered research to ensure software solutions align with operational workflows and warfighter needs. They gather insights through interviews, observations, and usability testing, translating findings into actionable design recommendations. UI/UX Researchers collaborate closely with developers and product teams to improve usability, reduce cognitive load, and enhance mission effectiveness. Their work ensures that digital tools are intuitive, efficient, and effective in operational environments.

Product Managers (PMs) – Define and prioritize software solutions that address critical operational challenges. They gather user requirements, manage backlogs, and align development efforts with mission objectives. Product Managers coordinate across cross-functional teams to ensure timely delivery of impactful capabilities. Their leadership ensures that software efforts remain focused, agile, and responsive to the evolving needs of the warfighter.

Chapter 4

Marine Coders Operating Model

“Talk is cheap. Show me the code.”

- Linus Torvalds

“War is ninety percent information.”

- Napoleon Bonaparte

Marine Coders operate within a hybrid software development framework, balancing structured collaboration with the Marine Corps Software Factory (MCSWF) and asynchronous, self-driven work. Given the part-time nature of the force, the operating model prioritizes high-impact development cycles, clear prioritization of projects, and streamlined workflows that maximize effectiveness within the constraints of approximately 40 drill periods and two weeks of annual training per Marine Coder.

Recruiting

The Marine Coders work closely with the MIU Talent Management Office and COMMSTRAT team for recruiting members of the team. In so doing, they will prioritize identifying Marines with a demonstrated interest in software development, a baseline of technical proficiency, and the maturity to operate independently within a mission-focused, self-directed environment.

Eligibility and Selection

- Reservists with industry experience in software development, platform engineering, data, or product management
- Active-duty Marines with technical aptitude and coding experience looking to transition to the Marine Corps Reserve
- Prospective candidates must apply through MIU recruitment channels

Pathways to Become a Marine Coder

- Reserve inter-unit transfer or joining MIU through talent board process
- Active Duty Transition – Marines transitioning from active service with technical skills can join via the Direct Affiliation Program (DAP) to MIU

- Lateral Entry from Civilian Sector – Industry professionals with prior Marine Corps service can return to uniform as Marine Coders.
- Internal Training & Upskilling – Marines can enhance their technical skills and transition into software development roles within MIU.

Screening

The MIU Talent Management Office will conduct the initial screening process for Marine Coders, evaluating candidates based on the aforementioned criteria as well as the practical administrative requirements necessary for onboarding.

Once a candidate has been screened and approved by Talent Management, the Marine Coders team will conduct a secondary assessment, focusing on technical proficiency and overall maturity, with maturity being the primary consideration.

Training

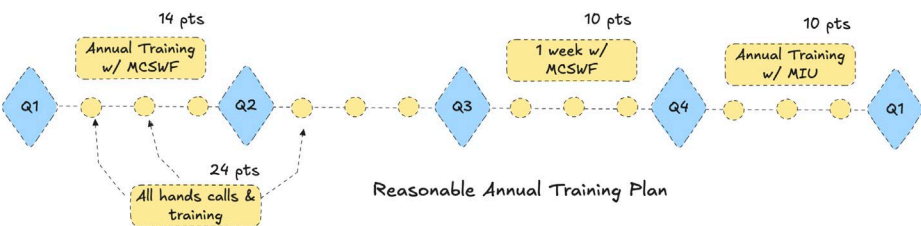
As members of the Marine Coders, our developers operate under a unique training model that emphasizes self-directed, asynchronous learning to maintain proficiency in their billet. Unlike other units, which may conduct structured training sessions, Marine Coders do not receive formalized training events from the unit. Instead, each Marine is individually responsible for identifying and addressing any skill gaps related to their billet requirements, leveraging online resources, peer collaboration, and real-world project experience to stay mission-ready. This decentralized approach ensures that Marines are continuously improving their technical expertise, adapting to emerging technologies, and remaining fully capable.

Employment

As with all reservists, Marine Coders are required to achieve a satisfactory year by earning at least 50 retirement points per anniversary year (not a calendar year). There are many ways to earn these 50 points, but a conventional means to do so is through a combination of the main Marine Coders events throughout the year, to include:

- Annual Training in support of MCSWF (14 points) - Once a year, the Marine Coders team will conduct a two-week on-site development sprint in support of MCSWF
- One-week offsite IDT in support of MCSWF (10 points) - Once a year, the Marine Coders team will conduct a one-week on-site development sprint in support of MCSWF
- One-week offsite IDT for MIU (10 points) - In general, plan on a one-week unit event in November for all-hands training and the Marine Corps ball
- One day a month unit calls and training (24 points) - Each month, the MIU hosts an all-hands call. The Marine Coders hosts training on these days as well, allowing participants to earn two points
- Each Friday, the Marine Coders team hosts office hours from 11am to 12am EST

While specific timelines vary year-to-year, a reasonable annual timeline for the Marine Coders team is as follows:



Collaboration with MCSWF and the Marine Corps

To ensure mission alignment, Marine Coders maintain structured engagement with MCSWF and other Marine Corps stakeholders through:

- Quarterly Planning Sessions – Marine Coders participate in roadmap planning and feature prioritization with MCSWF leadership to ensure efforts remain aligned with Force Design 2030 objectives
- Cross-Team Collaboration – Marine Coders act as technical resources to active-duty Marines at MCSWF, supporting them in modern software development, DevSecOps, and AI-driven warfighting applications

Scalability and Future Growth

As Marine Coders scale in numbers and capability, the operating model will evolve to support:

- Rotational Tasking Model – Aligning projects with seasonal availability of Reservists, allowing teams to cycle between development, mentorship, and operational support roles
- Rapid Surge Capacity – Marine Coders can be activated on short-term orders during crises, ensuring the Marine Corps has a ready reserve of digital warfighters who can rapidly respond to emergent threats

Scope of Marine Coders' Work

While the Marine Coders team represents a significant advancement in the Marine Corps' ability to deliver organic software capability, it is essential to understand the scope and scale at which this capability is

most effectively employed. Marine Coders are not intended to serve as a replacement for traditional acquisition pathways, nor are they structured to support large-scale, multi-year, enterprise-wide software systems. These types of efforts—often requiring sustained funding, lifecycle management, systems integration, and long-term support—remain the responsibility of the established acquisition community and Program Managers.

Rather, Marine Coders are most effectively applied to discrete, well-bounded software problems, particularly those with a clear operational or tactical need and a well-defined end user. The team excels when positioned as the primary force responsible for delivery, leveraging its agility and proximity to the warfighter to iterate rapidly and deliver minimum viable capabilities within compressed timeframes. In this capacity, Marine Coders can shorten the gap between user-identified need and functional capability, enabling a more responsive and adaptive approach to software in support of current operations.

This model aligns with the Marine Corps' broader emphasis on mission command, initiative, and adaptability. By focusing on problems that are small in scope but high in impact, Marine Coders are able to deliver solutions that address friction points across the force—whether in data collection, situational awareness, logistics visibility, or planning support. These solutions may not scale to the enterprise level, but they do scale to the needs of a battalion, a MEU, or a detachment—units whose effectiveness often hinges on timely access to tailored digital tools.

The value of Marine Coders is not in replacing formal acquisition pipelines, but in augmenting the Corps' ability to deliver capability at the tactical edge. When applied to the right problems—those that are immediate, bounded, and operationally relevant—Marine Coders serve as a force multiplier, enabling Marines to adapt the digital battlespace with the same mindset they bring to physical maneuver.

Chapter 5

Impact and Future Operations

“Measuring programming progress by lines of code is like measuring aircraft building progress by weight.”

- Bill Gates

“The most dangerous phrase: we’ve always done it this way.”

- Dr. Admiral Grace Hooper

Marine Coders have already begun to demonstrate meaningful impact across the force by delivering software capabilities that enhance mission effectiveness, accelerate decision-making, and reduce friction in both tactical and administrative environments. From building lightweight, deployable web applications to prototyping TAK plugins and automating critical workflows, Marine Coders have provided tangible, user-focused solutions that address operational needs in real time.

By embedding Marine software developers and technologists directly into problem-solving efforts, the Marine Corps gains not only functional tools, but also a responsive, agile capability that adapts to the pace of the modern battlefield. Unlike traditional acquisition cycles, which are often years in duration, Marine Coders are capable of delivering mission-relevant capabilities in weeks or even days. This ensures that warfighters receive tools when and where they need them—on timelines dictated by the mission, not by bureaucracy.

Looking forward, Marine Coders will continue to expand their support to the Marine Corps Software factory, delivering software in support of their priorities, emerging operational concepts, and evolving mission sets. Areas of continued and future focus include:

- Cross-Functional Team Integration – Working in close collaboration with Marine operators, planners, and acquisition professionals to co-create solutions that reflect tactical, operational, and strategic intent
- Platform Development and Application Hosting – Contributing to the development of a secure, Marine Corps-owned digital platform capable of hosting applications across classification levels and deployment environments. This foundational capability is critical to ensuring that Marine-developed software can be fielded rapidly, securely, and independently of enterprise constraints, supporting true agility and digital self-sufficiency

As the Marine Corps continues to modernize under the Force Design initiative, the demand for software that is secure, adaptable, and purpose-built for the warfighter will only grow. Marine Coders are uniquely positioned to meet this demand. Their ability to bridge operational insight with technical skill allows them to deliver high-impact solutions that scale from the fire team to the fleet.

The future operating environment requires a force that thinks, decides, and acts at the speed of relevance. Marine Coders ensure the Marine Corps is not only ready for that future—but is actively shaping it.

Conclusion

As the Marine Corps continues its ambitious evolution through the Force Design initiative, the critical role played by software cannot be overstated. The digital battlefield demands warriors who are proficient not only in traditional warfighting but also in modern software engineering practices. Marine Coders, drawn from the ranks of skilled reservists with extensive industry experience, serve as an essential bridge between innovative technology and operational effectiveness. Their unique combination of technical expertise, operational awareness, and adaptability enables the Marine Corps to swiftly respond to emerging challenges and exploit opportunities in real-time.

Ultimately, the Marine Coders are at the forefront of shaping a Marine Corps equipped to prevail in digitally contested environments. By continually developing and updating this guiding document, we ensure that current and future Marine Coders understand the significance of their role, the breadth of their responsibilities, and their profound impact on national security. As warfare evolves, Marine Coders will remain central to maintaining the Marine Corps' decisive technological advantage—ready to face and overcome the digital battlespace challenges of the 21st century.

Appendix

Technical Manager Billet Description

Software Engineer Billet Description

Data Analyst / AI/ML Specialist Billet Description

UI/UX Researcher Billet Description

Product Manager Billet Description

Core Skills for All Marine Coders

Billet Title

Technical Manager – Marine Coders Team, Marine Corps Reserve

Billet Description

The Technical Manager (Senior Software Developer) serves as the principal technical lead within the Marine Coders team, providing both hands-on software development and strategic technical guidance across multiple projects. This billet is responsible for shaping technical direction, mentoring junior developers, and ensuring the delivery of secure, scalable, and mission-aligned software solutions that directly support Marine Corps operational and administrative priorities.

As a senior member of a cross-functional team, the Technical Manager plays a key role in architecting systems, writing and reviewing code, and translating operational requirements into functional applications. They work in close coordination with product managers, UX designers, and stakeholders to ensure solutions are responsive to warfighter needs and are built with performance, usability, and security in mind.

This billet requires a high level of technical expertise, initiative, and leadership. It is best suited for Marines with extensive software development experience and the ability to operate independently in a remote, asynchronous, and fast-paced environment. Technical Managers are expected to uphold engineering excellence while helping shape the digital capability of the Marine Corps.

Primary Duties and Responsibilities

- Lead the technical design, development, and deployment of full-stack software applications
- Architect scalable, maintainable systems using modern development frameworks and tools
- Write, test, and review code to ensure performance, security, and adherence to best practices
- Provide technical mentorship and guidance to fellow developers and cross-functional teammates
- Collaborate with product managers, UX designers, and end users to refine requirements and deliver impactful solutions
- Support the integration of Marine Coders-developed software into operational workflows and Marine Corps platforms, including TAK and edge-deployable systems
- Participate in agile development practices, including sprint planning, backlog grooming, and retrospectives
- Maintain current knowledge of software engineering practices, DoD cyber requirements, and emerging technologies

Preferred Skills and Experience

- Proficiency in multiple modern programming languages and frameworks (e.g., React, Spring Boot, Python, Java, SQL, Git).
- Experience leading software development efforts or architecting complex systems in production environments
- Familiarity with tactical environments, military workflows, and digital modernization initiatives within the DoD
- Experience with TAK plugin development, secure mobile or edge computing applications, and geospatial data formats (GeoPackage, KML, MBTiles)
- Familiarity with DevSecOps principles, CI/CD pipeline management, and infrastructure-as-code tools
- Strong communication skills and the ability to translate between technical and operational stakeholders

Billet Title

Software Developer – Marine Coders Team, Marine Corps Reserve

Billet Description

The Marine Corps Reserve Software Developer serves as a key member of a cross-functional development team responsible for the design, development, deployment, and maintenance of software solutions in direct support of Marine Corps operational and administrative needs. This billet supports the mission of the Marine Coders and other Reserve-led digital initiatives by providing technically proficient Marines who can develop mission-relevant applications in an agile and responsive manner.

Software developers are expected to contribute across the entire software development lifecycle, including requirements gathering, system design, code implementation, testing, and deployment. Applications may include tactical decision aids, internal tools, web applications, data dashboards, and integrations with existing Marine Corps systems, including TAK-based platforms.

This billet requires asynchronous, self-directed training and participation, with Marines responsible for maintaining the skills and certifications necessary to perform in a modern software environment. Collaboration with other developers, product owners, and end-users is essential, and Marines in this billet must demonstrate maturity, operational awareness, and a continuous learning mindset.

Primary Duties and Responsibilities

- Develop and maintain software applications using modern programming languages and frameworks (e.g., React, Spring Boot, Python, JavaScript, SQL)
- Contribute to tactical applications such as ATAK plugins or edge-deployable web applications
- Participate in agile development practices, including sprint planning, code reviews, and user feedback cycles
- Collaborate with active duty and Reserve components to understand operational requirements and translate them into functional software
- Conduct testing, debugging, documentation, and deployment of software solutions
- Continuously train to remain proficient in relevant technologies and software development practices

Preferred Skills and Experience

- Proficiency in one or more of the following: React, Spring Boot, Python, Java, Android/Kotlin, SQL, Docker, Git
- Familiarity with tactical environments, military workflows, or digital modernization efforts within the DoD
- Experience with geospatial data formats (GeoPackage, KML, MBTiles) and platforms such as TAK is highly desirable
- Ability to work independently and as part of a remote, distributed team

Billet Title

Data Analyst / AI/ML Specialist – Marine Coders Team, Marine Corps Reserve

Billet Description

The Data Analyst / AI/ML Specialist supports the integration of data-driven decision-making and artificial intelligence capabilities into Marine Corps operations and institutional functions. These Marines serve on cross-functional teams and are responsible for the collection, analysis, visualization, and modeling of data to inform warfighter needs, improve operational efficiency, and enhance command and control (C2) capabilities.

Marines in this billet are expected to develop analytical tools and machine learning models tailored to tactical, operational, or administrative use cases. This may include building dashboards, data pipelines, recommendation systems, or AI-enabled decision support tools for planning, logistics, intelligence, and personnel systems.

This is a technical, self-directed, and impact-oriented billet, requiring Marines to train asynchronously and apply industry-standard methods in data science and AI/ML. Collaboration with software developers, subject matter experts, and operational units is critical to mission success.

Primary Duties and Responsibilities

- Design and execute data analyses to support operational planning, readiness, and performance optimization.
- Build and maintain data pipelines, ETL processes, and automated analytics workflows.
- Develop machine learning models and AI-enabled tools to solve targeted Marine Corps problems.
- Create intuitive data visualizations, dashboards, and decision aids using tools like Power BI, Tableau, Plotly, or open-source libraries.
- Collaborate with end users to translate operational challenges into analytical questions and actionable outputs.
- Integrate with software development teams (e.g., Marine Coders) to embed analytics and models into functional applications.
- Stay abreast of emerging trends in data science, AI/ML, and DoD data initiatives (e.g., JADC2, Project Maven, ADVANA).

Preferred Skills and Experience

- Proficiency in Python, R, SQL, or other data science languages and tools.
- Experience with pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, or similar frameworks.
- Familiarity with data engineering tools (e.g., Airflow, DBT, Spark), cloud platforms, or tactical edge computing.
- Ability to analyze and work with structured and unstructured military data sets, including logistics, intelligence, and personnel data.
- Prior experience in military operations, operations research, or supporting decision-making with data is highly desirable.

Billet Title

UI/UX Researcher – Marine Coders Team, Marine Corps Reserve

Billet Description

The UI/UX Researcher serves as the primary advocate for the end user within the Marine Coders team, ensuring that software solutions are intuitive, usable, and tailored to the operational needs of Marines. This billet plays a critical role in bridging the gap between users and developers by conducting research that informs design decisions across the full product lifecycle.

UI/UX Researchers are responsible for gathering, analyzing, and synthesizing user insights through interviews, usability testing, field observation, and other research methods. They work closely with product managers, designers, and developers to translate research findings into actionable recommendations that improve the functionality, accessibility, and overall effectiveness of Marine Coders-developed software.

This billet requires a deep understanding of user-centered design principles, operational empathy, and the ability to function within an agile, distributed development team. The ideal candidate is mission-driven, collaborative, and capable of conducting independent, high-quality research that improves outcomes at the tactical and operational levels.

Primary Duties and Responsibilities

- Plan and execute user research efforts, including interviews, contextual inquiries, usability tests, and surveys
- Synthesize research findings into clear, actionable insights to inform design and development decisions
- Collaborate with product managers, designers, and developers to align software features with user needs and workflows
- Create user personas, journey maps, task flows, and other UX artifacts to communicate findings and design direction
- Validate design concepts and prototypes through iterative testing with representative users
- Advocate for accessibility, usability, and mission effectiveness in all user-facing aspects of the application
- Contribute to the creation of style guides and interface standards to ensure consistent and user-friendly experiences
- Remain current with best practices in UX research, design thinking, and human-centered design methodologies

Preferred Skills and Experience

- Experience conducting UX research using both qualitative and quantitative methods
- Familiarity with design tools such as Figma, Adobe XD, Sketch, Miro, or similar platforms
- Ability to produce clear research documentation and communicate insights to technical and non-technical audiences.
- Understanding of agile software development and working within cross-functional teams
- Experience designing for users in complex or high-stakes environments (e.g., military, medical, industrial)
- Strong interpersonal skills and the ability to build trust and rapport with end users and stakeholders
- Self-directed and able to contribute meaningfully in a distributed, asynchronous work environment

Billet Title

Product Manager – Marine Coders Team, Marine Corps Reserve

Billet Description

The Product Manager serves as the strategic and operational lead for software product development within the Marine Coders team. This billet is responsible for identifying and prioritizing operational needs, translating them into actionable product requirements, and guiding cross-functional teams through the delivery of effective, user-centered software solutions in support of the warfighter.

Product Managers define product vision, manage roadmaps, and ensure alignment between Marine Corps operational challenges and technical efforts. They serve as the connective tissue between stakeholders, developers, designers, and end users—ensuring that each product delivered by the team is focused, mission-relevant, and built to enhance operational effectiveness.

This role requires strong operational awareness, technical fluency, and communication skills, as well as the ability to lead in a dynamic, distributed, and often asynchronous team environment. Product Managers must be comfortable operating in ambiguity and making informed decisions based on user feedback, strategic guidance, and tactical realities.

Primary Duties and Responsibilities

- Define product goals, roadmaps, and success criteria in collaboration with stakeholders and users
- Translate operational needs into clear, prioritized user stories and technical requirements
- Lead agile ceremonies including sprint planning, backlog refinement, and retrospectives
- Coordinate closely with developers, UX researchers, designers, and data analysts to ensure cohesive product delivery
- Gather and synthesize user feedback to inform product direction and ongoing iteration
- Communicate product vision and progress to stakeholders, both technical and operational
- Manage scope, timelines, and prioritization to ensure delivery of mission-relevant outcomes
- Maintain awareness of Marine Corps modernization efforts and digital priorities to inform product strategy

Preferred Skills and Experience

- Experience leading software or digital product development efforts using agile or lean methodologies
- Strong skills in requirements gathering, user story writing, and backlog management
- Familiarity with tools such as Jira, Trello, GitHub Projects, or similar platforms
- Understanding of software development processes, technical architecture, and design constraints
- Experience working directly with military end users or in support of operational units
- Ability to communicate effectively across technical and non-technical audiences
- Demonstrated ability to manage competing priorities and deliver results in resource-constrained environments

Core Skills for All Team Members

While the individual roles each bring distinct capabilities to the Marine Coders team, there is a set of foundational skills that all team members should possess. These shared competencies enable collaboration, increase interoperability, and ensure a baseline level of technical fluency across the team.

Foundational Technical Skills

Git & Version Control

- Understand basic Git operations: clone, pull, commit, push, branching
- Comfortable using GitHub or GitLab for code and documentation management
- Enables collaboration, version control, and integration with CI/CD workflows

Setting Up a Local Development Environment

- Ability to install and configure an IDE (e.g., VS Code, IntelliJ, PyCharm)
- Set up local environments using tools like Docker, Python virtual environments, or Node/npm
- Supports local development, debugging, and testing of software

Command Line Fundamentals

- Navigate directories, manage files, and run scripts via terminal or shell
- Useful across development, data analysis, and deployment tasks

Understanding of APIs and JSON

- Familiarity with RESTful APIs, including how to test endpoints
- Able to read and work with JSON-formatted data
- Enables integration across systems and services

Basic Data Literacy

- Understand structured data formats such as CSV, JSON, and SQL
- Able to read and interpret simple datasets and write/read basic queries
- Supports better decision-making, debugging, and development

Software Development Lifecycle Awareness

- Understands agile concepts like sprints, user stories, and minimum viable products (MVPs)
- Familiar with testing, code review, and deployment cycles

Security & Operational Discipline

- Basic awareness of DoD cybersecurity principles
- Understands the importance of secure data handling, access control, and compliance
- Critical for developing software in military environments

Documentation & Communication

- Able to write and read clear README files, usage guides, and process documentation
- Comfortable with collaborative documentation tools (e.g., Markdown, Google Docs)
- Promotes clarity, continuity, and effective collaboration in a distributed team

Deployment Awareness

- Understands the general process of how code moves from development to production
- Familiar with CI/CD principles and configuration basics (e.g., YAML pipeline files)
- Enables more seamless collaboration with DevSecOps personnel

UX and User Empathy

- Understands the value of usability, clarity, and responsive interfaces
- Appreciates the importance of user feedback and intuitive design
- Improves coordination with UX designers and leads to more mission-effective tools

Recommended Tools and Platforms for Familiarity

- GitHub or GitLab (for version control and collaboration)
- Visual Studio Code (or another preferred IDE)
- Docker (basic usage for building and running containers)
- Markdown (for documentation)
- Postman (for API testing and validation)
- Terminal / Command Line
- Common team communication tools (e.g., Slack, Microsoft Teams, Discord, Matrix)